

Hodge Deligne Numbers for Hypersurfaces in Toric Varieties

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1 Mike Meeting 03 Aug

Mike took us through some of the basic ideas. It was a long meeting, but there were a lot of ideas, and so this is a writeup for some of the ideas from that meeting.

Any text in purple indicates either a question from myself or an incomplete thought.

1.1 Definition of Hodge-Deligne Numbers and Some Basic Properties

Let X be a variety over $\mathbb{C}$ of dimension¹ n . The *Hodge-Deligne numbers* denoted $e^{p,q}(X)$ for $0 \leq p, q \leq n$ are a collection of integers satisfying properties soon to be described. Moreover the polynomial $e_X(x, y) := \sum_{0 \leq p, q \leq n} e^{p,q}(X) x^p y^q$ is called the *Hodge-Deligne polynomial*. The Hodge-Deligne numbers and polynomial satisfy the following properties

- The Hodge-Deligne polynomial is symmetric in x, y , $e_X(x, y) = e_X(y, x)$. In other words, the Hodge-Deligne numbers are symmetric in p, q , $e^{p,q}(X) = e^{q,p}$.
- If X is reducible then the Hodge-Deligne polynomial decomposes additively over the components: if $X = Y \amalg Z$ (presumably these have to be closed subsets?) then $e_X = e_Y + e_Z$. (A variety is by definition irreducible, so, perhaps the most general setting is for a separated, finite type over $\mathbb{C}$ -scheme? Or maybe this property is supposed to be for closed subvarieties?)
- If X is a product variety then its Hodge-Deligne polynomial decomposes multiplicatively: if $X = Y \times_{\mathbb{C}} Z$ then $e_X = e_Y e_Z$.
- If X is projective and smooth (over $\mathbb{C}$) then recall that the hodge numbers are the dimensions of the $\mathbb{C}$ -vector spaces given by cohomology over the sheaf of differentials

$$h^{p,q}(X) = \dim_{\mathbb{C}} H^q(X, \Omega_X^p).$$

¹A variety is defined to be separated, irreducible, finite type k -scheme. I am trying to remember why we have that a variety has a well-defined dimension.

In this setting, the Hodge-Deligne numbers are related to the Hodge numbers in the following way

$$e^{p,q}(X) = (-1)^{p+q} \cdot h^{p,q}(X).$$

1.2 Examples

Example 1 (A point). Let $X = \{*\} = \text{Spec}(\mathbb{C})$. We have that X is smooth and projective. The following Hodge calculation will give that $e_X(x, y) = 1$.

Since X is affine $H^q(X, \Omega_X^p)$ is non-vanishing only for $q = 0$. Recall that $\Omega_X^p = 0$ for all $p > 0$ (I have forgotten why this is true, but it follows because X is a point. I think it should be easy to build the module of differentials in this case once one remembers the definition and construction) and so $h^{p,q} = 0$ for all $p, q > 0$. The only possible non-zero Hodge number we may have is for $p = q = 0$. We have the following

$$\begin{aligned} h^{0,0}(X) &:= H^0(X, \Omega_X^0) \\ &= \dim_{\mathbb{C}} H^0(X, \mathcal{O}_X) && \text{By definition} \\ &= \dim_{\mathbb{C}} \Gamma(X, \mathcal{O}_X) && \text{By the cohomology axioms} \\ &= \dim_{\mathbb{C}} \mathbb{C} && X = \text{Spec}(\mathbb{C}) \\ &= 1. \end{aligned}$$

Overall, we have that $h^{0,0}(X) = 1$ and $h^{p,q}(X) = 0$ for all other p, q . The claimed result follows.

Example 2 (Union of Points). It follows from the previous example and the decomposition property of e_X that a disjoint union of n points has Hodge-Deligne polynomial $e_X(x, y) = n$.

Example 3 ($\mathbb{P}_{\mathbb{C}}^1$). Performing a similar Hodge computation to the example of the point one can compute that

$$e_{\mathbb{P}_{\mathbb{C}}^1} = 1 + xy.$$

This might be a worthwhile computation. $\mathbb{P}^1$ is covered by two affine patches and so we only need to compute for $p < 2$. We need to recall how to compute the sheaf of differentials however, but I think it should be nonvanishing only degree 0, 1.

Example 4 ($\mathbb{A}_{\mathbb{C}}^1$). We can decompose $\mathbb{P}^1 = \mathbb{A}^1 \amalg \{*\}$ the affine line with a “point at infinity”. “point at infinity.” Since $\mathbb{A}^1$ is an open subset and $\{*\}$ is a closed point we have

$$e_{\mathbb{A}_{\mathbb{C}}^1}(x, y) = e_{\mathbb{P}_{\mathbb{C}}^1} - e_{\{*\}} = xy.$$

I'm a bit confused by this example, because the affine line is affine, and so we should expect it to only

have have cohomology in degree 0. That is to say, I would expect that this should be a polynomial only in y . The Hodge-Deligne numbers agree with the Hodge numbers only when X is projective – $\mathbb{A}^1$ is not projective, so we're gucci.

Example 5 ($\mathbb{A}_{\mathbb{C}}^n$). Using the product property of the Hodge-Deligne polynomial we have

$$E_{\mathbb{A}^n} = (xy)^n.$$

From this example, we can read off the hodge numbers without needing to do any Hodge computations.

Example 6 (Algebraic Torii). We can decompose the affine line $\mathbb{A}^1 = \mathbb{C}^* \amalg \{*\}$ and so using the decomposition property of the Hodge-Deligne polynomial we have

$$e_{\mathbb{C}^*} = e_{\mathbb{A}_{\mathbb{C}}^1} + e_{\{*\}} = xy - 1.$$

Then, using the product property we further have that

$$e_{(\mathbb{C}^*)^n} = (e_{\mathbb{C}^*})^n = (xy - 1)^n.$$

Example 7 ($\mathbb{P}_{\mathbb{C}}^n$). Recall that we can decompose $\mathbb{P}^n = \mathbb{C}^n \amalg \mathbb{P}^{n-1}$ and then we can find the Hodge-Deligne polynomial of $\mathbb{P}^n$ inductively. Consider,

$$\begin{aligned} e_{\mathbb{P}^1} &= 1 + xy \\ e_{\mathbb{P}^2} &= e_{\mathbb{C}^2} + e_{\mathbb{P}^1} = (xy)^2 + (1 + xy) \\ e_{\mathbb{P}^3} &= e_{\mathbb{C}^3} + e_{\mathbb{P}^2} = (xy)^3 + (1 + xy + (xy)^2) \\ &\dots \\ e_{\mathbb{P}^n} &= 1 + (xy) + (xy)^2 + \dots + (xy)^n. \end{aligned}$$

2 Procedure for Hypersurfaces in Toric Varieties.

we should go back and fill out some of the preamble theory here

Suppose we have a full dimension polytope Δ . We can decompose our toric variety

$$P_{\Delta} = \amalg_{\Gamma' \leq \Delta} T_{\Gamma'}$$

where $T_{\Gamma'} = (\mathbb{C}^*)^{\dim \Gamma'}$. And so we can compute the Hodge-Deligne polynomial of a toric variety using the decomposition property. Does each subcone really just give a torus in a toric variety? Or

is it somehow sufficient to compute the Hodge-Deligne polynomial on a dense open?

Let $Z_{\Gamma'} = Z \cap T_{\Gamma'}$. We have the following formula

$$e^{pq}(\bar{Z}) = e^{pq}(Z) + \sum_{\Gamma' < \Delta} e^{pq}(Z_{\Gamma'}).$$

Note that a consequence of this formula is that for $p = n$ or $q = n$, $e^{pq}(\bar{Z})$ is determined by $e^{pq}(Z)$. I think this should follow from the dimension property of Hodge-Deligne numbers, and each $e^{pq}(Z_{\Gamma'})$ has dimension strictly less than n . Is this reasoning correct?

We also have the following formulae

$$e^{pq}(Z) = e^{p+1, q+1}(T) = \begin{cases} 0 & p \neq q \\ (-1)^{n+p+1} \binom{n}{p+1} & p = q \end{cases} \quad (1)$$

for $p + q \geq n$. And,

$$\sum_{q=0}^{n-1} e^{pq}(Z) = (-1)^p \left((-1)^{n+1} \binom{n}{p+1} - \sum_{k \geq 1} (-1)^k \binom{n+1}{p+k+1} \ell^*(k\Delta) \right). \quad (*)$$

where ℓ^* denotes the number of interior lattice points of the given polytope. Is this last formula actually correct? Are there conditions on p, q for when this formula holds?

I think the algorithm is then as follows: we iteratively compute the Hodge-Deligne numbers starting from the smallest cone, and we only need to use formula (*) to add the numbers in degree n , is that correct?

2.1 Examples

Example 8 (Hypersurface in $\mathbb{P}^1 \times \mathbb{P}^1$). Recall that the projective toric variety $\mathbb{P}^1 \times \mathbb{P}^1$ is given by the polytope $\Delta = \text{ConvexHull}((1, 1), (1, -1), (-1, 1), (-1, -1))$, the reflexive square with the origin as the interior lattice point. We use the recursive procedure described above to compute the Hodge-Deligne numbers for the hypersurface defined by

$$f(x, y) = xy + \frac{x}{y} + \frac{1}{xy} + \frac{y}{x}.$$

Note that the Newton polytope of this Laurent polynomials is in fact Δ . First we determine that this polynomial is non-degenerate with respect to Δ . Let us label each of the facets (edges) of Δ as follows; $\Gamma_1 = \text{ConvexHull}((-1, 1), (1, 1))$ as the “top face”, $\Gamma_2 = \text{ConvexHull}((1, 1), (1, -1))$ as the

“right face”, and proceeding clockwise. The facet² polynomials are then as follows

$$\begin{aligned} f_{\Gamma_1} &= \frac{y}{x} + xy \\ f_{\Gamma_2} &= xy + \frac{x}{y} \\ f_{\Gamma_3} &= \frac{x}{y} + \frac{1}{xy} \\ f_{\Gamma_4} &= \frac{1}{xy} + \frac{y}{x}. \end{aligned}$$

We show that $Z(f_{\Gamma_1})$ is smooth. The partial derivatives are

$$\frac{\partial f_{\Gamma_1}}{\partial x} = -\frac{y}{x^2} + y \quad \frac{\partial f_{\Gamma_1}}{\partial y} = \frac{1}{x} + x.$$

$\partial f_{\Gamma_1}/\partial y = 0$ has solutions in $(\mathbb{C}^*)^2$ with x -coordinate $x = \pm 1$ whereas $\partial f_{\Gamma_1}/\partial x = 0$ implies $x = \pm i$. This system has no simultaneous solutions in $(\mathbb{C}^*)^2$ and so $Z(f_{\Gamma_1})$ is smooth. A similar calculation will give that each $Z(f_{\Gamma_i})$ is smooth and thus f is indeed non-degenerate with respect to Δ .

Next we compute the Hodge-Deligne numbers for $Z = Z(f)$. Since we are in $n = 2$, in principle, we have nonvanishing Hodge-Deligne numbers for $p, q \leq n$; However formula (1) above implies that for $p + q \geq 2$ the only nonvanishing Hodge number is $e^{11}(Z) = (-1)^{2+1+1} \binom{2}{1+1} = 1$.

Next we compute the contributions from $\sum_{\Gamma' < \Delta} e^{pq}(Z_{\Gamma'})$. Let us first consider $\Gamma' = \Gamma_1$. By definition this is $Z(f_{\Gamma_1}) \cap T_{\Gamma_1} \cong Z(f_{\Gamma_1}) \cap \mathbb{C}^*$. Recall $f_{\Gamma_1} = y/x + xy = y(1/x + x)$. The primitive normal vector corresponding to Γ_1 is $(0, 1)$, and so intersecting with T_{Γ_1} algebraically corresponds to sending $y \mapsto 1$ (I'm not this part is articulated correctly; I think restricting to this cone corresponds to restricting to an open patch where we “divide by y ”, similar to the distinguished patches in projective space? Is that correct?). And so we search for solution in $\mathbb{C}^*$ to $1/x + x = 0$. The solution set is $x = \pm i$; topologically this is the disjoint union of two points in $\mathbb{C}^*$. It follows then from the union-sum property of Hodge-Deligne polynomials and from the Hodge-Deligne polynomial of a point found above that $e_{Z_{\Gamma_1}} = 2$. A very similar argument yields that $e_{Z_{\Gamma'}} = 2$ for all $\Gamma < \Delta$. And so $e^{00}(\bar{Z}) = 8$.

The last Hodge-Deligne numebr to compute is $e^{10}(Z)$, we compute this using the formula (*). We have

²I think in the general setting the vertex monomials are always vacuously smooth since I think they do not even have solutions in $(\mathbb{C}^*)^2$. This argument about the general setting should be checked, but one can certainly verify here that the vertex monomials are smooth.

$$\begin{aligned}
e^{10}(Z) + e^{11}(Z) &= (-1)^1 \left[(-1)^{2+1} \binom{2}{1+1} - \sum_{k \geq 1} (-1)^k \binom{2+1}{1+k+1} \ell^*(k\Delta) \right] \\
e^{10}(Z) + 1 &= - \left[(-1) - (-1) \binom{3}{3} (1) \right] \\
e^{10}(Z) &= -1.
\end{aligned}$$

By p, q symmetry of the Hodge-Deligne numbers, we³ also have $e^{01}(Z) = -1$.

Overall, we have found the Hodge-Deligne polynomial for f as

$$e_Z(x, y) = 8 - x - y + xy.$$

Is there a good way to check if this makes sense?

³As an aside, interestingly, this portion of the calculation seems to not have depended on the choice of hypersurface at all, is this correct?

3 graveyard

Example 9 ($\mathbb{P}^1 \times \mathbb{P}^1$). Here we compute the Hodge-Deligne numbers for $Z = \mathbb{P}^1 \times \mathbb{P}^1$. Recall that for this toric variety we have $\Delta = \text{ConvexHull}((1, 1), (1, -1), (-1, 1), (-1, -1))$. Since $n = 2$ we have that there are only four Hodge-Deligne numbers to compute.

We start with $p = 1, q = 1$. This is the only case for which $p + q \geq 2$ and we have

$$e^{11}(Z) = e^{2,2}(T) = (-1)^{2+1+1} \binom{2}{1+1} = 1.$$

Next we compute e^{01} . For this we use $(*)$. We have

$$\begin{aligned} e^{10} + e^{11} &= (-1)^1 \left[(-1)^{2+1} \binom{2}{1+1} - \sum_{k \geq 1} (-1)^k \binom{1+1}{1+k+1} \ell^*(k\Delta) \right] \\ e^{10} + 1 &= (-1) \left[(-1) - \left((-1)^1 \binom{3}{3} \cdot 1 \right) \right] \\ e^{10} &= -1. \end{aligned}$$

Then by p, q symmetry we have that $e^{01} = -1$ also.

Lastly, we compute e^{00} . We can do this by recurring and computing the Hodge-Deligne number for a unit ray. Recall that the toric variety corresponding to a single ray is the affine line $\mathbb{C}^1$. We computed earlier that this has Hodge-Deligne polynomial $e_{\mathbb{C}^1} = xy$. Meaning that the Hodge numbers for $\mathbb{C}^1$ are all zero except for $p = 1, q = 1$.